In [ ]:
```
EXPERIMENT 5 – DATA VISUALIZATION WITH MATPLOTLIB
PIN NO.: 241UDS0109

AIM:-
To visualize and analyze a dataset using Matplotlib line charts, scatter plots, and subplots for better
interpretation of trends and patterns.
```

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df_241UDS0109 = pd.read_csv("Experiment_5_241UDS0109_Renewable_Energy.csv")
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17 entries, 0 to 16
Data columns (total 13 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Year                            17 non-null     int64
 1   Solar_Generation_GWh            17 non-null     float64
 2   Wind_Generation_GWh             17 non-null     float64
 3   Hydro_Generation_GWh            17 non-null     float64
 4   Electricity_Price_INR           17 non-null     float64
 5   Coal_Price_USD                  17 non-null     float64
 6   Inflation_Rate                  17 non-null     float64
 7   Renewable_Subsidy_Crore         17 non-null     int64
 8   Clean_Energy_Households_Million 17 non-null     float64
 9   Renewable_Share_%               17 non-null     float64
 10  Demand_Growth_%                 17 non-null     float64
 11  Electricity_to_Coal_Ratio       17 non-null     float64
 12  Subsidy_per_GWh                 17 non-null     float64
dtypes: float64(11), int64(2)
memory usage: 1.9 KB
```

In [ ]:
```
## Step 2: Display the Dataset
```

In [2]:
```
df_241UDS0109.head()
```

Out[2]: Columns 1–8 of 14

	Year	Solar_Generation_GWh	Wind_Generation_GWh	Hydro_Generation_GWh	Electricity_Price_INR	Coal_Price_USD	Inflation_Rate
0	2010	18.4	72.5	110.2	4.8	78.2	10.4
1	2011	22.1	78.3	112.4	4.9	105.4	8.9
2	2012	27.5	84.6	115.1	5.1	98.6	9.3
3	2013	34.2	91.8	118.6	5.3	101.2	10.1
4	2014	43.7	99.4	121.3	5.6	85.7	6.5

Columns 9–14 of 14 · first column repeated

	Renewable_Subsidy_Crore	Clean_Energy_Households_Million	Renewable_Share_%	Demand_Growth_%	Electricity_to_Coal_Ratio	Subsidy_per_GWh
0	820	12.5	8.2	6.1	0.0614	170.8
1	910	14.1	9.1	7.3	0.0465	185.7
2	1040	16.2	10.4	8.6	0.0517	203.9
3	1190	18.8	11.7	9.2	0.0524	224.5
4	1370	21.3	13.2	10.1	0.0653	244.6

In [3]:
```
df_241UDS0109
```

Out[3]: Columns 1–8 of 14

	Year	Solar_Generation_GWh	Wind_Generation_GWh	Hydro_Generation_GWh	Electricity_Price_INR	Coal_Price_USD	Inflation_Rate
0	2010	18.4	72.5	110.2	4.8	78.2	10.4
1	2011	22.1	78.3	112.4	4.9	105.4	8.9
2	2012	27.5	84.6	115.1	5.1	98.6	9.3
3	2013	34.2	91.8	118.6	5.3	101.2	10.1
4	2014	43.7	99.4	121.3	5.6	85.7	6.5
5	2015	55.8	110.2	124.8	5.8	49.6	5.0
6	2016	71.4	121.7	128.2	5.7	48.1	4.7
7	2017	89.6	135.9	132.7	6.0	55.3	3.6
8	2018	112.3	149.2	137.4	6.2	71.6	3.4
9	2019	139.8	164.7	142.1	6.5	63.8	4.8
10	2020	165.2	178.3	146.8	6.7	45.1	6.2
11	2021	198.5	195.6	151.2	7.0	69.4	5.5
12	2022	235.7	214.9	155.7	7.3	96.8	6.7
13	2023	278.4	236.1	160.3	7.6	81.3	5.4
14	2024	325.6	258.7	165.1	7.9	79.5	4.6
15	2025	379.2	282.4	170.6	8.2	67.1	3.9
16	2026	438.5	309.8	176.2	8.5	92.4	3.5

Columns 9–14 of 14 · first column repeated

	Renewable_Subsidy_Crore	Clean_Energy_Households_Million	Renewable_Share_%	Demand_Growth_%	Electricity_to_Coal_Ratio	Subsidy_per_GWh
0	820	12.5	8.2	6.1	0.0614	170.8
1	910	14.1	9.1	7.3	0.0465	185.7
2	1040	16.2	10.4	8.6	0.0517	203.9
3	1190	18.8	11.7	9.2	0.0524	224.5
4	1370	21.3	13.2	10.1	0.0653	244.6
5	1510	24.7	14.8	12.0	0.1169	260.3
6	1635	28.2	16.1	13.8	0.1185	286.8
7	1790	32.5	17.9	15.2	0.1085	298.3
8	1955	37.8	19.8	17.4	0.0866	315.2
9	2180	43.2	21.6	18.9	0.1019	335.4
10	2370	49.1	23.5	19.4	0.1486	353.7
11	2590	55.6	25.4	21.8	0.1009	370.0
12	2815	62.4	27.6	23.2	0.0754	385.6
13	3050	69.8	29.7	24.6	0.0935	401.3
14	3290	77.2	31.9	26.5	0.0994	416.5
15	3540	85.6	34.1	28.7	0.1222	431.7
16	3820	94.3	36.4	30.1	0.0920	449.4

In [4]:
```
df_241UDS0109.columns
```

Out[4]:
```
Index(['Year', 'Solar_Generation_GWh', 'Wind_Generation_GWh',
       'Hydro_Generation_GWh', 'Electricity_Price_INR', 'Coal_Price_USD',
       'Inflation_Rate', 'Renewable_Subsidy_Crore',
       'Clean_Energy_Households_Million', 'Renewable_Share_%',
       'Demand_Growth_%', 'Electricity_to_Coal_Ratio', 'Subsidy_per_GWh'],
      dtype='object')
```

In [8]:
```
df_241UDS0109["Year"] = pd.to_numeric(df_241UDS0109["Year"])
```

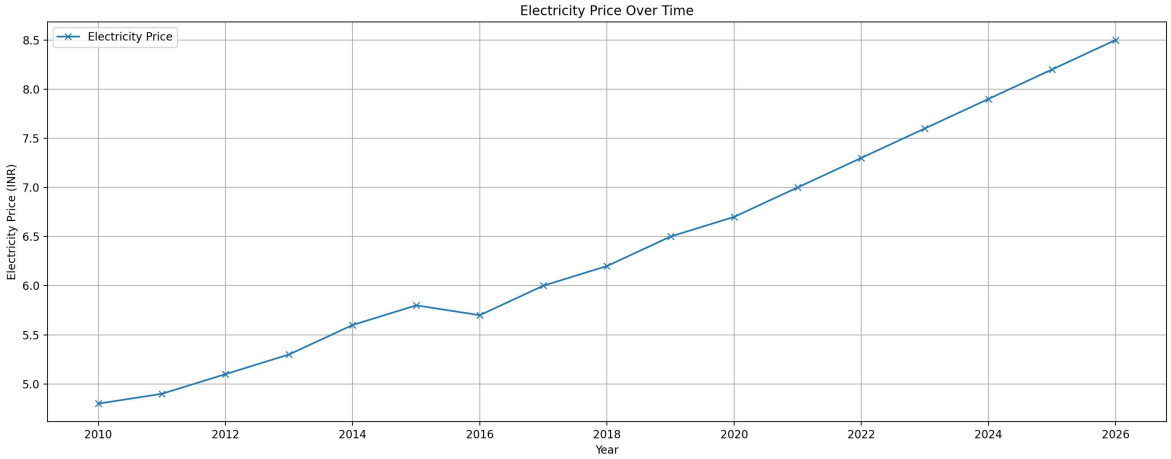
In [ ]:
```
## Step 3: Visualize Data Using Various Plots
### 1. Electricity Price Over Time
```

In [9]:
```
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0109["Year"],
    df_241UDS0109["Electricity_Price_INR"],
    marker="x",
    label="Electricity Price"
)

plt.title("Electricity Price Over Time")
plt.xlabel("Year")
plt.ylabel("Electricity Price (INR)")

plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



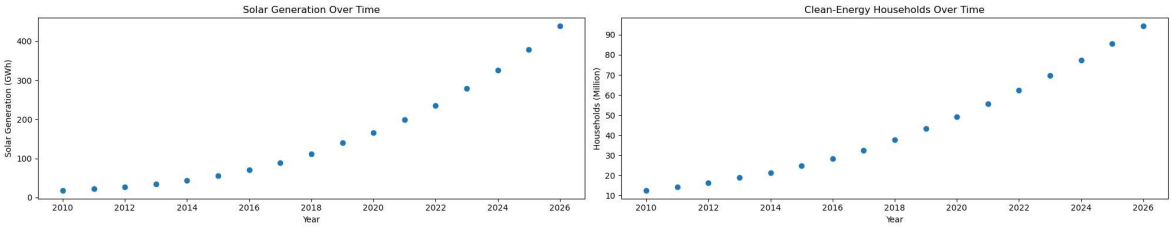
In [ ]:
```
### 2. Subplots — Renewable Energy Generation and Clean-Energy Households
```

In [10]:
```
fig, axes = plt.subplots(1, 2, figsize=(20, 4))

# Solar generation
axes[0].scatter(
    df_241UDS0109["Year"],
    df_241UDS0109["Solar_Generation_GWh"]
)
axes[0].set_title("Solar Generation Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Solar Generation (GWh)")

# Clean-energy households
axes[1].scatter(
    df_241UDS0109["Year"],
    df_241UDS0109["Clean_Energy_Households_Million"]
)
axes[1].set_title("Clean-Energy Households Over Time")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Households (Million)")

plt.tight_layout()
plt.show()
```



In [ ]:
```
### 3. Comparing Solar, Wind and Hydro Generation
```

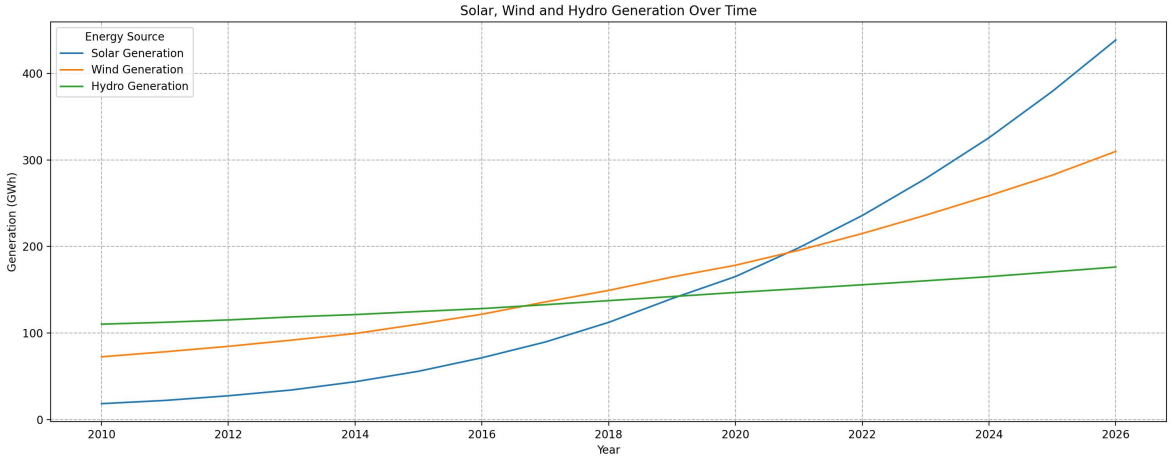
In [11]:

```
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0109["Year"],
    df_241UDS0109["Solar_Generation_GWh"],
    label="Solar Generation"
)
plt.plot(
    df_241UDS0109["Year"],
    df_241UDS0109["Wind_Generation_GWh"],
    label="Wind Generation"
)
plt.plot(
    df_241UDS0109["Year"],
    df_241UDS0109["Hydro_Generation_GWh"],
    label="Hydro Generation"
)

plt.title("Solar, Wind and Hydro Generation Over Time")
plt.xlabel("Year")
plt.ylabel("Generation (GWh)")

plt.tight_layout()
plt.legend(loc="upper left", title="Energy Source")
plt.grid(True, linestyle="--")
plt.show()
```



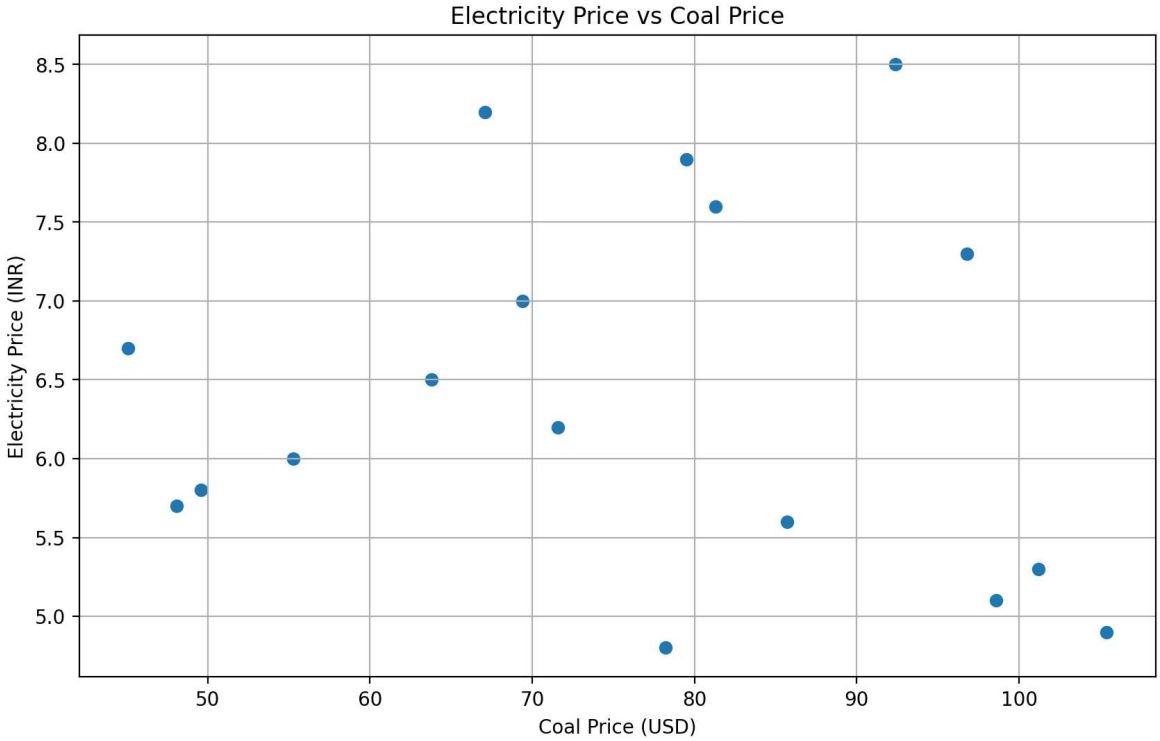
In [ ]:

```
## Step 4: Electricity Price vs Coal Price
```

In [12]:
```
plt.figure(figsize=(10, 6), dpi=200)

plt.scatter(
    df_241UDS0109["Coal_Price_USD"],
    df_241UDS0109["Electricity_Price_INR"]
)

plt.title("Electricity Price vs Coal Price")
plt.xlabel("Coal Price (USD)")
plt.ylabel("Electricity Price (INR)")
plt.grid(True)
plt.show()
```



In [ ]:
```
## Step 5: Renewable-Energy Subsidy Over Time
```

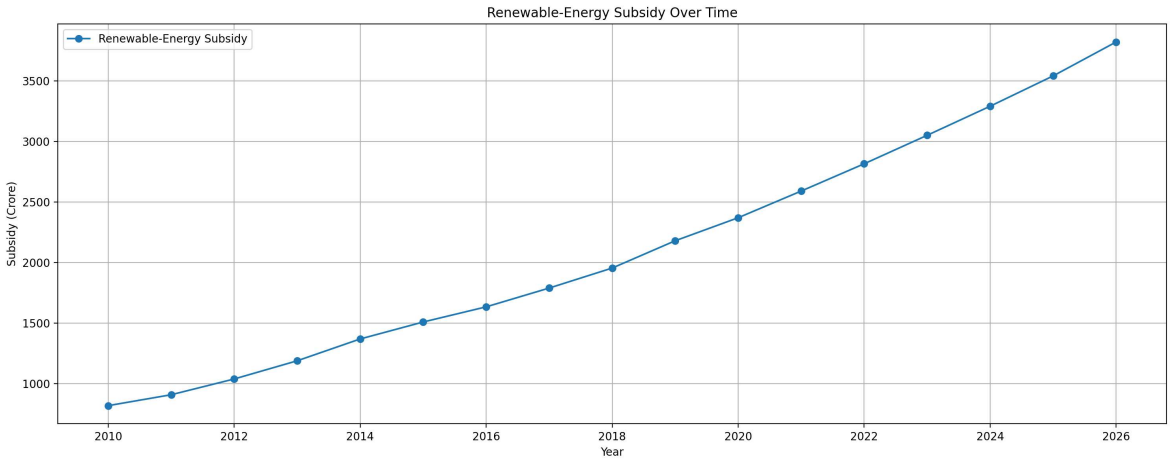
In [13]:

```
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0109["Year"],
    df_241UDS0109["Renewable_Subsidy_Crore"],
    marker="o",
    label="Renewable-Energy Subsidy"
)

plt.title("Renewable-Energy Subsidy Over Time")
plt.xlabel("Year")
plt.ylabel("Subsidy (Crore)")

plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



In [ ]:

RESULT:-
The plots show electricity-price trends, solar generation and clean-energy households over time, comparisons of solar/wind/hydro generation, electricity price versus coal price, and renewable-energy subsidy trends.